Course Name: Geographic Information System

Week 04 - Assignment 04 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

1mm on a map drawn at a scale of 1:50,000 represent _____ on the ground.

a) 500 cm
b) 50 cm
c) 50 m
d) 5 m

Answer: (c)

Detailed Solution:

A map scale of 1:50,000 means 1 unit on the map represents 50,000 of the same units on the ground. Therefore, 1 mm on the map = 50,000 mm = 50 m on the ground.

QUESTION 2

How is a city most likely to be represented on a 1:25,000 scale map?

a) As a single point
b) As a line
c) As a collection of points
d) As a collection of points, lines and areas

Answer: (d)

Detailed Solution:

On a 1:25,000 large-scale map, a city is shown in greater detail using points, lines, and areas to represent roads, buildings, rivers, and boundaries.

QUESTION 3

The _____ projection is an example of a cylindrical projection.

a) State Plane Coordinate System (SPCS)
b) Azimuthal
c) Mercator

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d) All of the above

Answer: (c)

Detailed Solution:

The Mercator projection is a type of cylindrical projection because it projects the Earth's surface onto a cylinder. It is widely used for navigation.

QUESTION 4

The prime meridian runs through _____.

a) Equator
b) Greenwich
c) Greenland
d) Moscow
e)

Answer: (b)

Detailed Solution:

The Prime Meridian passes through Greenwich, England, and is defined as 0° longitude. It is the reference line for measuring east and west longitudes.

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QUESTION 5

A small-scale map would show

a) A larger geographic area than a large-scale map
b) A smaller geographic area than a large-scale map
c) The same geographic area as a large-scale map, just at a smaller resolution
d) None of the above

Answer: (a)

Detailed Solution:

A small-scale map covers a larger geographic area but with less detail compared to a large-scale map. Examples include world or country maps.

QUESTION 6

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_____ projection is accurate in the mid-latitude zone.

a) Mercator
b) Conic
c) Azimuthal
d) Cylinder

Answer: (b)

Detailed Solution:

Conic projections are most accurate in the mid-latitude regions because distortion is minimized along the standard parallels. They are commonly used for regional maps.

QUESTION 7

Conformal maps could preserve _____ of a map property.

a) Area
b) Distance
c) Direction
d) Shape

Answer: (d)

Detailed Solution:

Conformal maps preserve the shape of small areas and maintain local angles, although area and distance may become distorted.

QUESTION 8

Actual area of Greenland is _____ than/to Africa, and this is due to Mercator distortion.

a) Much larger
b) Much smaller
c) Equal
d) None of the above

Answer: (b)

Detailed Solution:

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Greenland appears much larger on the Mercator projection but in reality, its actual area is much smaller than Africa. This difference is caused by distortion near the poles.

QUESTION 9

Which of the following represents positions as X, Y and Z coordinates (3D)?

a) Geographic coordinate system
b) Rectangular coordinate system
c) Planar coordinate system
d) Geocentric coordinate system

Answer: (d)

Detailed Solution:

A geocentric coordinate system represents locations using X, Y, and Z coordinates in three-dimensional space, measured from the Earth's centre.

QUESTION 10

Georeferencing using mailing address is also known as _____.

a) Geocoding
b) Direct georeferencing
c) Continuous georeferencing
d) Relative georeferencing

Answer: (a)

Detailed Solution:

Assigning geographic coordinates using a mailing address is called geocoding. It converts addresses into spatial locations that can be mapped in GIS.

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